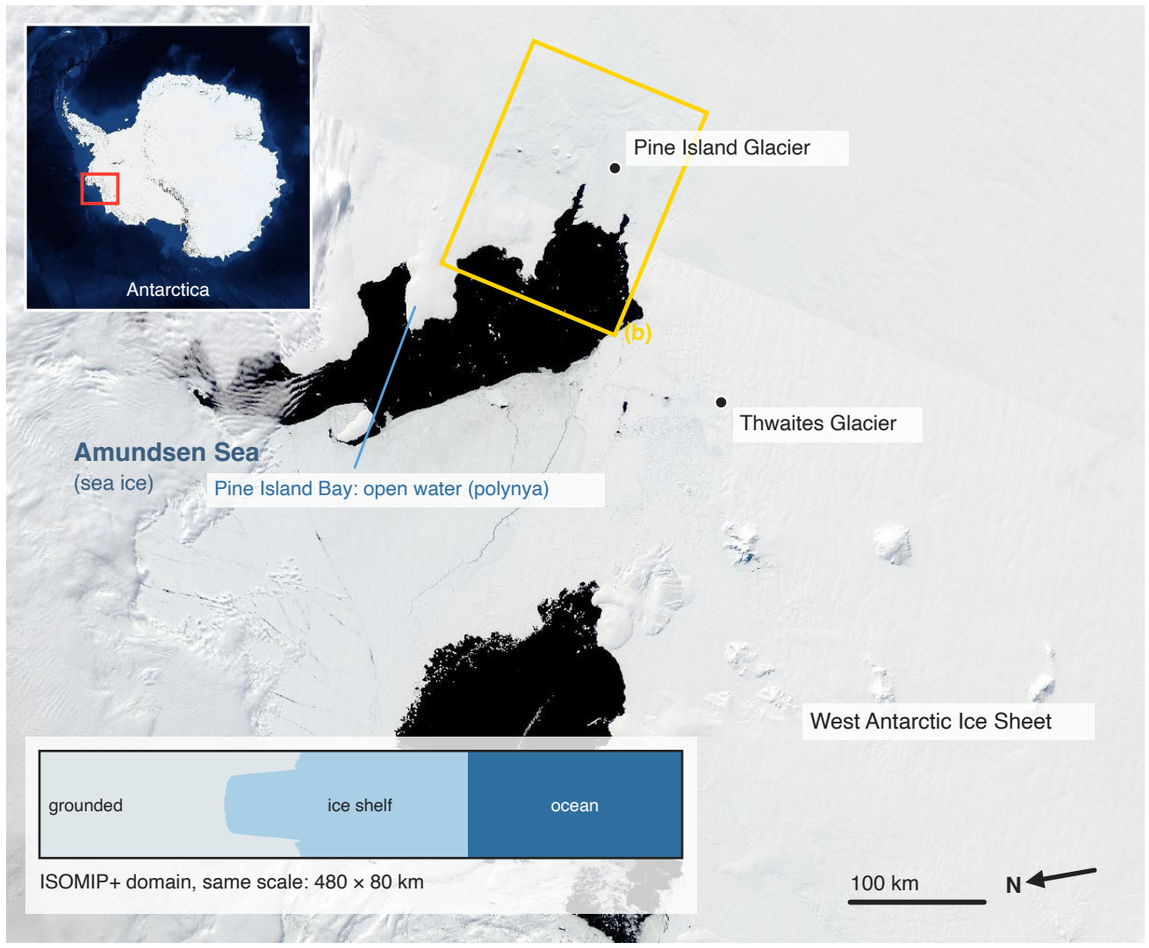
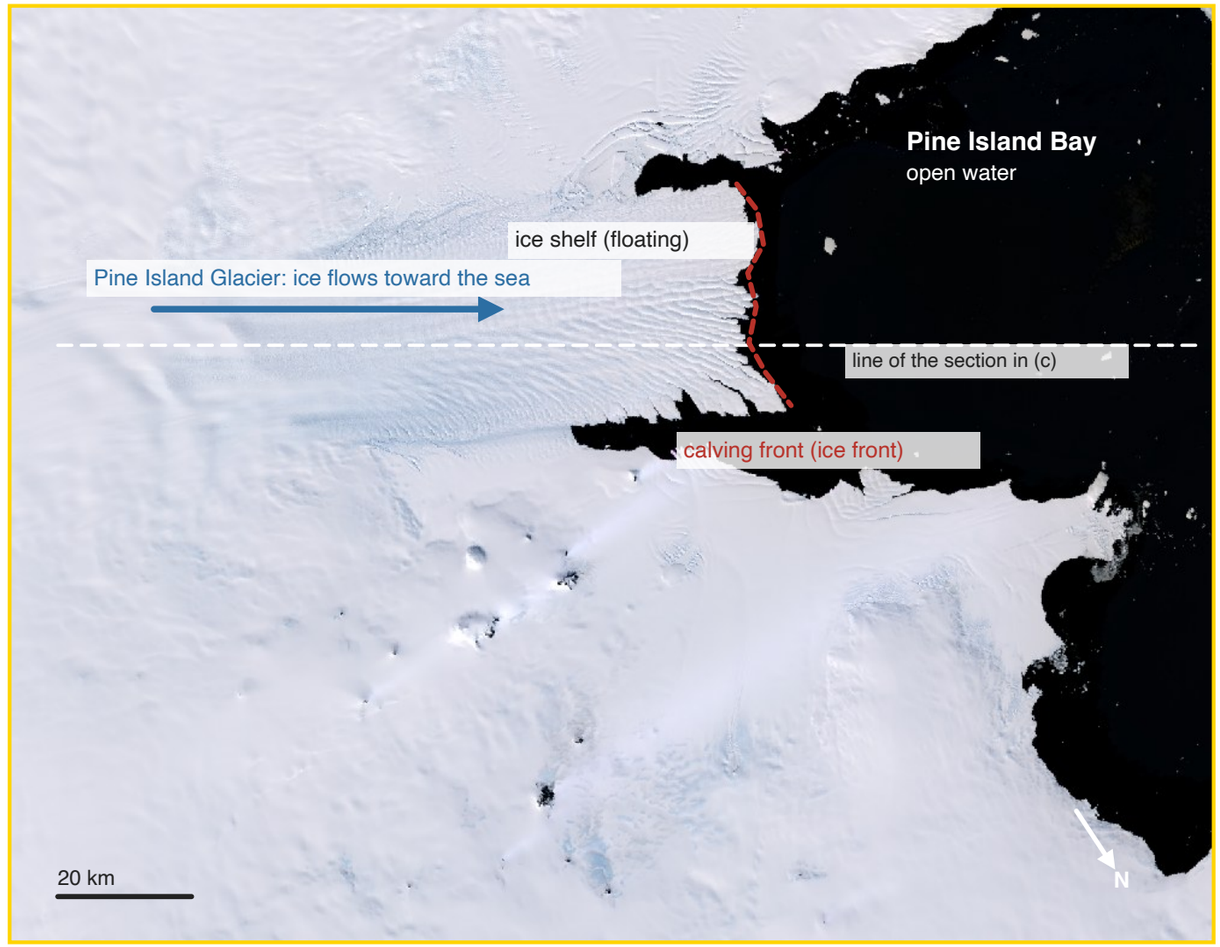
(a) Amundsen Sea Embayment, West Antarctica

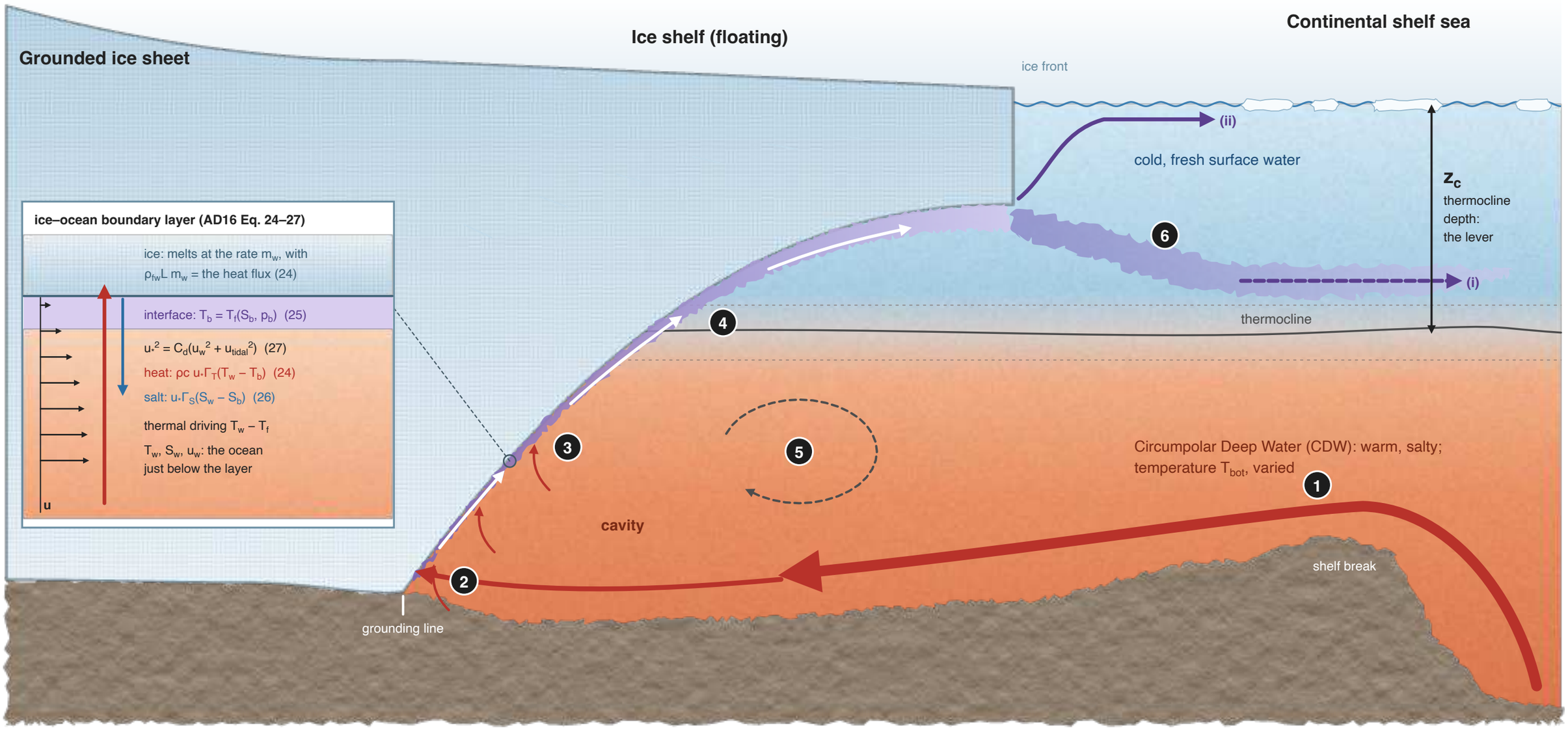


(b) Pine Island Glacier ice shelf, Landsat 9, 30 m



Imagery: (a) MODIS Terra true colour, 1 Dec 2025, NASA Worldview/GIBS; inset: NASA Blue Marble with shaded relief and bathymetry. (b) Landsat 9 OLI-2, 1 Dec 2025, courtesy of the U.S. Geological Survey; rotated so that the ice flows to the right.

(c) Processes beneath an ice shelf (a section along the ice flow)



(c) is a schematic: not to scale (the cavity, a few hundred metres thick, is several tens of kilometres long). Process 7, rotation, is drawn in Figure 0d.

1. warm, salty Circumpolar Deep Water flows onto the continental shelf at depth, below the thermocline
2. the CDW reaches the grounding line, where the ice base is deepest and the freezing point lowest
3. basal melting: heat and salt cross a thin boundary layer at the ice base (inset)
4. the meltwater is buoyant and rises as a plume along the sloping ice base
5. the melt drives an overturning: inflow at depth, outflow along the ice base
6. the meltwater leaves the cavity: (i) it detaches near its neutral-buoyancy depth, or (ii) it rises to the surface
7. Earth's rotation steers the circulation into currents along the cavity walls (Figure 0d)